Introduction to Machine Learning and Deep Learning

ML is the study of algorithms that enable computers to learn patterns from data and make decisions like a human might.

ML methods fall into three main types:

- Supervised learning:

  Models learn from *labeled* data (known input and output). For instance, a spam filter learns to label emails as "spam" or "not spam."

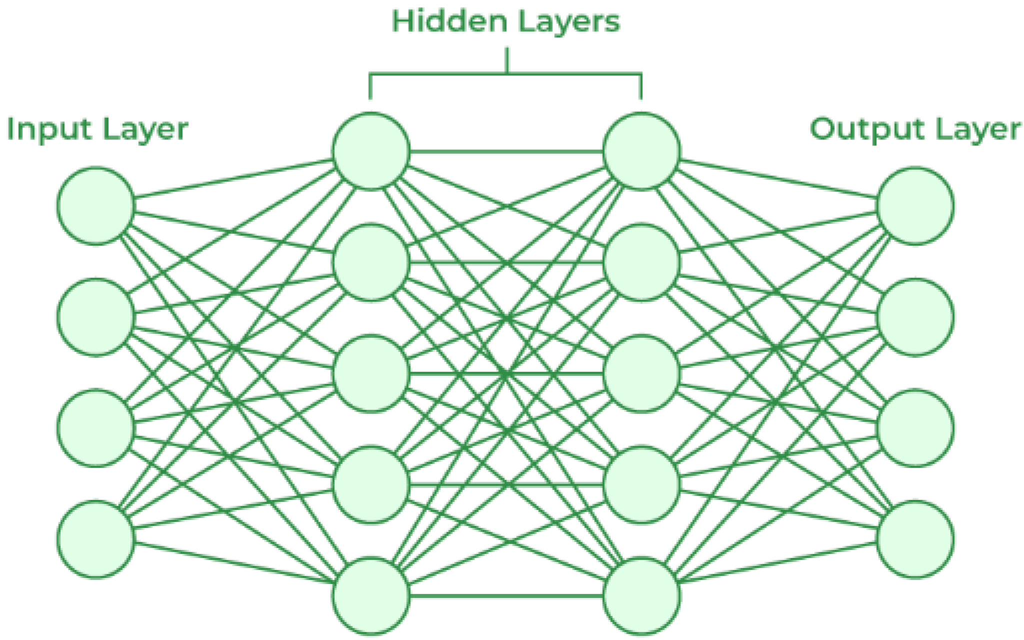
- Unsupervised learning:

  Models find hidden structures or patterns in *unlabeled* data and group/ categories them accordingly . For example, unsupervised learning algorithms group similar customer profiles without predefined labels.

- Reinforcement learning: Models learn by trial and error through rewards or penalties (like teaching a game-playing AI by rewarding wins).

Difference between ML and DL

1. Traditional ML often requires manual feature engineering. That is, human experts must define which input patterns matter. lets say, to recognize a face, we would have to tell the computer exactly what a face looks. It can be the positions of eyes, nose, mouth, etc..
2. Deep learning overcomes this by automatically learning relevant features from raw data.For example: In facial recognition, a deep model can analyze multiple face images and *learn* that certain pixel patterns correspond to eyes, noses, etc., without being explicitly programmed with those features. This mimics the human ability to recognize faces quickly without thinking about individual features.
3. Deep learning is a subset of ML where models use many layered neural networks to learn from data.
4. A deep model with many hidden layers can capture complex patterns by building up simple features into more abstract concepts.
5. Because of this, deep learning excels at tasks like image and speech recognition, natural language processing, and more, often outperforming traditional ML on these problems



Artificial Neural Networks (ANNs)

An artificial neural network (ANN) is a collection of connected nodes organized in layers, resembling a simplified brain(logical animal brain).It has an input layer , one or more hidden layers, and an output layer.

Each connection has a *weight* called a multiplier and each neuron may have a *bias* .

When data flows through the network, each neuron computes a weighted sum of its inputs plus bias eg: $z = W \cdot x + b$. That sum is passed through an activation function, which decides the neuron's output.

Forward Propagation

Forward propagation is the process of sending input data through the network to compute an output. It works as follows:

1. Input Layer: The raw data (e.g. image pixels, audio samples, or numeric features) enter the input layer, one value per input neuron.

2. Hidden Layers: Each neuron in a hidden layer takes all its inputs, computes a weighted sum plus bias (such as $z = W \cdot x + b$), and applies an activation function. For example, an image pixel value might be multiplied by a weight, added to a bias, and then passed through ReLU (which outputs the sum if positive, or 0 if negative). The output of one

layer becomes the input to the next. Early layers learn simple features (edges, colors), while deeper layers learn complex patterns (shapes, object parts

3. Output Layer: The final layer produces the network's prediction. For classification, it might use a softmax activation to output probabilities for each class. The network then predicts the class with highest probability.

Backward Propagation

This is the learning phase where the network adjusts its weights based on the error of its output. The process is:

- Compute Loss: After forward propagation, the network's output is compared to the true label using a loss function by for example mean squared error or cross-entropy, giving a numerical error value.

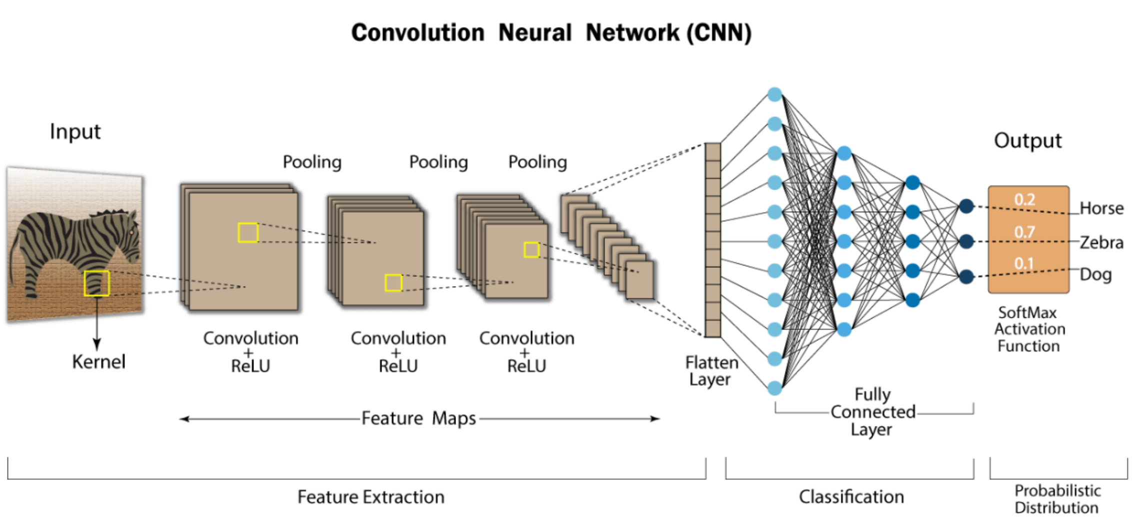
- Propagate Error Backwards: Backpropagation uses the chain rule to determine how each weight and bias contributed to the total error. It computes the gradient of the loss with respect to each parameter by going *backward* from the output layer to the input layer.

- Update Weights: The network then adjusts each weight and bias slightly to reduce the error. For example, if changing a weight decreases the loss, it is adjusted in that direction; otherwise it is adjusted in the opposite direction.

In effect, backpropagation tells the model "how wrong it was" and how to fix it.

Deep Learning Architectures and Algorithms

- **Convolutional Neural Networks**



CNNs are designed for grid-like data, especially images. They use convolutional layers with small filters that slide across the image to detect features like edges or textures and pooling layers that downsample features. Because of this, CNNs recognize patterns regardless of their position in the image. CNNs have achieved groundbreaking performance in image classification and computer vision In practice, CNNs power facial recognition, self-driving car, obstacle detection, medical image analysis. For example, a CNN trained on many labeled photos can learn to identify cats, dogs, or faces with high accuracy.

- **Recurrent Neural Networks (RNNs)**

RNNs are used for sequential data like text or time series. Unlike feedforward networks, RNNs maintain an internal "memory" of previous inputs, allowing them to use past context when processing new data. This means the output at each step depends on both the current input and the hidden state from previous steps. Common RNN variants include LSTMs and GRUs, which help learn long-term dependencies. RNNs are widely used in language and speech tasks: for instance, predicting the next word in a sentence, translating languages, or analyzing stock prices over time. An example: an RNN trained on English-French sentence pairs can learn to translate a whole sentence by reading words sequentially and remembering context.

Key Terms

- Attention Mechanisms: Attention is a technique that allows a model to focus on the most relevant parts of its input when making a decision. In sequence models, an attention layer weighs different input tokens to capture their importance. For example, when translating a sentence, attention helps the model decide which source words are most relevant to each output word.
- Self-attention computes attention weights *within* a sequence, so the model learns relationships between all token pairs at once. Attention is crucial in handling long-range dependencies: it enables the model to “attend” to relevant words no matter how far apart they are. As IBM explains, attention lets the network dynamically predict the importance of each element in the sequence.

- Transformer Models: Transformers are a family of models built almost entirely on self-attention layers. Instead of processing sequence data one step at a time (like RNNs), transformers look at the entire input sequence together. As Nvidia describes, a transformer is a neural network that “learns context and thus meaning by tracking relationships in sequential data”. Transformers use multi-headed self-attention to detect subtle influences between all parts of the input (for example, which earlier words should influence the current word). This parallel processing makes them very powerful and efficient to train on large datasets. Transformer-based models like BERT or GPT are now the state-of-the-art in NLP. They excel at tasks like translation, question-answering, and text generation. For instance, Google Translate and language-chatbots (e.g. ChatGPT) use transformer architectures to understand and produce human-like text.

- Generative Adversarial Networks (GANs): A GAN consists of two competing neural networks – a *generator* and a *discriminator*. The generator creates synthetic data (e.g. fake images), while the discriminator evaluates whether inputs are real or generated. As training progresses, the generator learns to produce increasingly realistic data that can “fool” the discriminator. For example, a GAN trained on photos of faces can generate new, photorealistic faces that never existed. GANs are widely used for image synthesis, style transfer, and data augmentation. A real-life example: converting a photo of a horse into a zebra, or generating high-quality artwork and textures.

- Activation function : Introduce non-linearity into the network also decides whether the neuron can contribute to the next layer.

  Different types of activation function :

  a. Step function : activate a neuron if it is some value if not it is not activities. The drawback is if you have multiple classes(non binary classification)

b. Linear function: activation is proportional to input this way it gives a range of activation (analog values).if more then one activates or fires we could take the maximum of it too.the problem here is the
c. Sigmoid function : non linear so good for data with non linear non binary nature so we can stack layers. Has analog outputs. Unlike linear function the output is going to be between 0 to 1 where input (x) can be -infinity to +infinity

Drawbacks of Deep Learning

While powerful, deep learning has some limitations:

- Large Data Requirements: Deep models need huge amounts of data to train effectively. For instance, recognizing cats vs. dogs might require millions of labeled images. Collecting and labeling such data can be difficult and expensive.

- High Computational Cost: Training deep neural networks is computationally intensive. They often require specialized hardware like GPUs or TPUs to train in a reasonable time For example, CNNs for image tasks usually run on GPUs because their many matrix operations are too slow on regular CPUs.

- Lack of Interpretability: Deep models are often “black boxes.” It can be hard to understand why a deep network made a specific decision. This is a concern in critical fields (like medicine) where explanations are important.

- Overfitting and Bias: If not enough data is available, or if the data is biased, deep models can overfit or learn those biases, reducing their real-world reliability.

Despite these drawbacks, deep learning’s ability to automatically learn complex features and achieve high accuracy has made it a central technology in modern AI. By using neural networks with many layers and leveraging large datasets, deep learning algorithms like CNNs, RNNs, attention-based transformers, and GANs power a wide range of applications from image recognition to language translation, often mimicking how humans perceive and generate information.

Sources: Definitions and explanations are drawn from authoritative AI and deep learning resources, including IBM and industry blog references. Each cited source provides detailed insights into the concepts above.